

**Problem 61.** A sequence  $a_0, a_1, a_2, \dots$  of positive integers satisfies the following recurrence relation:

$$a_{n+1} = \begin{cases} \frac{1}{3}a_n, & \text{if } 3 \mid a_n, \\ 2a_n + 1, & \text{otherwise,} \end{cases}$$

for all integers  $n \geq 0$ .

Show that, as  $n$  increases to infinity, one of the following cases must hold:

Case 1: Either the sequence  $\{a_n\}$  increases to infinity, or

Case 2:  $\{a_n\}$  ends up in an alternating cycle  $1, 3, 1, 3, 1, \dots$ .

**Problem 62.** Find all triples  $(a, b, c)$  of positive integers such that

$$\begin{cases} a^2 + b^2 = n \operatorname{lcm}(a, b) + n^2, \\ b^2 + c^2 = n \operatorname{lcm}(b, c) + n^2, \\ c^2 + a^2 = n \operatorname{lcm}(c, a) + n^2, \end{cases}$$

for some positive integer  $n$ .

**Problem 63.** Let  $k > 1$  be a positive integer. Find the smallest positive integer  $n$  such that some cells of an  $n \times n$  grid of unit squares can be coloured black in such a way that each row and each column contains exactly  $k$  black cells and no two black cells have a common side or a common vertex.

**Problem 64.** Samuel and Mandy play the following game. Samuel starts by drawing a random triangle on a piece of paper. Mandy then draws a line on the same paper that goes through the midpoint of one of the sides of the medial triangle of the triangle. Then Samuel adds another line that also goes through the midpoint of the same side of the medial triangle of the triangle. These two lines divide the triangle into four pieces. Samuel gets the piece with maximum area (or one of those with maximum area) and the piece with minimum area (or one of those with minimum area), while Mandy gets the other two pieces. The player whose total area is bigger wins. Does either of the players have a winning strategy, and if so, who has it?